

EDUCATION-NEWS CONSULT – DODOWA, ACCRA

MATHEMATICS - MARKING SCHEME

OBJECTIVES

1. D	6. B	11. C	16.A	21. C	26. A	31. D	36. C
2. C	7. B	12. B	17. C	22. C	27. D	32. A	37. C
3. D	8. D	13. C	18. C	23. B	28. A	33. B	38. A
4. D	9. A	14. C	19. A	24. C	29. D	34. B	39. B
5. D	10. C	15. A	20. C	25. A	30. C	35. B	40. A

SECTION B

1. a. Let the total profit be P

Sum of ratio = 3+5=8

Share of A = $\frac{3}{8}P$ M$\frac{1}{2}$

Share of B = $\frac{5}{8}P$ M$\frac{1}{2}$

If B received GH¢3000 more than A

$\Rightarrow \frac{5}{8}P - \frac{3}{8}P = 3000$ M1

$\Rightarrow \frac{5P-3P}{8} = 3000$ M1

$\Rightarrow \frac{2P}{8} = 3000$ M1

$\Rightarrow \frac{P}{4} = 3000$

$\Rightarrow P = 3000 \times 4$

$P = 12000$ A1

Therefore total profit is GH¢12000

b. Given that $x = \binom{4}{5}, y = \binom{0}{-3}$ *and* $z = \binom{-2}{7}$, find $2(x + 3y + 5z)$

$2(x + 3y + 5z) \Rightarrow \quad 2\left[\binom{4}{5} + 3\binom{0}{-3} + 5\binom{-2}{7}\right]$ M1

$\Rightarrow \quad 2\left[\binom{4}{5} + \binom{0}{-9} + \binom{-10}{35}\right]$ M$\frac{1}{2}$

$\Rightarrow \quad 2\binom{-6}{31} = \binom{-12}{62}$ M1A1

c. $2x + 40 = 5x - 80$ (vertically opposite angles are equal) M1

$40 + 80 = 5x - 2x$ M$\frac{1}{2}$

$120 = 3x$

$\frac{120}{3} = \frac{3x}{3}$ M1

$x = 40$ A1

$5y - 15 = 2y + 30$ (vertically opposite angles are equal) M$\frac{1}{2}$

$5y - 2x = 30 + 15$ M$\frac{1}{2}$

$3x = 45$

$\frac{3x}{3} = \frac{45}{3}$ M1

$x = 15$ A1

$(x^2 + y^2) = (40^2 + 15^2)$ M1

$= 1600+225$

$= 1825$ A1

2. a. $\frac{1}{5}(2 - y) + 1 \leq \frac{1}{2}(3y - 4)$

L.C.M = 10

$10 \times \frac{1}{5}(2 - y) + 1 \times 10 \leq 10 \times \frac{1}{2}(3y - 4)$ M1

$2(2 - y) + 10 \leq 5(3y - 4)$ M$\frac{1}{2}$

$4 - 2y + 10 \leq 15y - 20$ M$\frac{1}{2}$

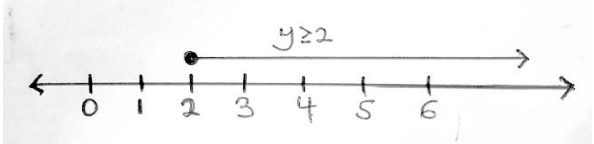
$-2y - 15y \leq -20 - 10 - 4$ M$\frac{1}{2}$

$-17y \leq -34$

$\frac{-17y}{-17} \leq \frac{-34}{-17}$ M$\frac{1}{2}$

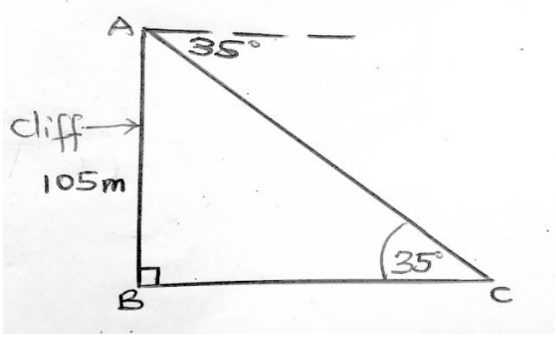
$y \geq 2$

Truth set = $\{y: y \geq 2\}$ A1



A1

b.



BC represents the distance of the boat from the foot to the cliff

$Tan\ 35 = \frac{105}{|BC|}$ M1

$|BC| = \frac{105}{Tan\ 35}$ M1

But Tan 35 = 0.613

$|BC| = \frac{105}{0.613}$ M1

$= 171.29\ m$

$|BC| = 171\ m$ to the nearest whole meter A1

c. Number of sides (n) = 5

Sum of interior angles $= (n-2)180$

$= (5-2)180$

$= (3)180$

$= 540°$ M1

$9x - 18 + 9x - 18 + 9x - 18 + 9x - 18 + 9x - 18 = 540$ M1

$45x - 90 = 540$ M$\frac{1}{2}$

$45x = 540 + 90$ M$\frac{1}{2}$

$45x = 630$

$\frac{45x}{45} = \frac{630}{45}$ M1

$x = 14$ A2

3. a. slope = $\frac{2}{3}$

Point = (-4, 5)

Equation of a line $\Rightarrow y - y_1 = m(x - x_1)$

$\Rightarrow y - 5 = \frac{2}{3}(x - -4)$ M1

$\Rightarrow y - 5 = \frac{2}{3}(x + 4)$

$\Rightarrow 3y - 15 = 2(x + 4)$ M$\frac{1}{2}$

$\Rightarrow 3y - 15 = 2x + 8$ M$\frac{1}{2}$

$\Rightarrow 3y = 2x + 8 + 15$

$3y = 2x + 23$ or $-2x + 3y - 23 = 0$ A2

b. (i) Total number of pupils = 50

Pupils who speak French only = $\frac{1}{10} \times 50 = 5$ pupils M$\frac{1}{2}$

Remainder = $50 - 5 = 45$ pupils M$\frac{1}{2}$

Pupils who speak English only = $\frac{4}{5} \times 45 = 36$ pupils $\quad M\frac{1}{2}$

Let x represent pupils speak both French and English

$\Rightarrow x +$ French only + English only $= 50$

$\Rightarrow x + 5 + 36 = 50 \quad M\frac{1}{2}$

$x = 50 - 41$

$x = 9 \quad A1$

Therefore 9 pupils speak both French and English

ii. Pupils who speak English only = 36 pupils $\quad A1$

c.. $\frac{2(m+n)-3(m-3n)}{(m-3n)(m+n)} \quad M1$

$\frac{2m+2n-3m+9n}{m(m+n)-3n(m+n)} \quad M1$

$\frac{2m-3m+2n+9n}{m^2+mn-3mn-3n^2} \quad M1$

$\frac{-m+11n}{m^2-2mn-3n^2} \quad A1$

But $m = 3 \; amd \; n = -2$

$\frac{-m+11n}{m^2-2mn-3n^2} \Rightarrow \frac{-3+11(-2)}{3^2-2(3)(-2)-3(-2)^2} \quad M1$

$\frac{-3-22}{9+12-12} = \frac{-25}{9} \quad M1A1$

4. i. Let x represent his monthly income.

Rent $\frac{1}{4}x \quad M\frac{1}{2}$

Remaining income$= x - \frac{1}{4}x = \frac{3x}{4} \quad M\frac{1}{2}$

$\Rightarrow$ Food $= \frac{1}{3} \times \frac{3x}{4} = \frac{3x}{12} = \frac{x}{4} \quad M\frac{1}{2}$

What still remained $= \frac{3x}{4} - \frac{x}{4} = \frac{2x}{4} = \frac{x}{2} \quad M\frac{1}{2}$

$\Rightarrow$ Clothes $= \frac{1}{8} \times \frac{x}{2} = \frac{x}{16} \quad M\frac{1}{2}$

Rent + Food + Clothes $= \frac{x}{4} + \frac{x}{4} + \frac{x}{16} \quad M\frac{1}{2}$

$= \frac{4x+4x+x}{16} \quad M\frac{1}{2}$

$= \frac{9x}{16} \quad A\frac{1}{2}$

Remaining salary $= x - \frac{9x}{16} \quad M\frac{1}{2}$

$= \frac{16x-9x}{16} \quad M\frac{1}{2}$

$= \frac{7x}{16} \quad A\frac{1}{2}$

But remaining salary = GH₵2100

$\Rightarrow \frac{7x}{16} = 2100 \quad M1$

$\Rightarrow 7x = 2100 \times 16$

$\Rightarrow 7x = 33600$ $\quad M\frac{1}{2}$

$x = 4800$

Therefore his total monthly income = GH₵4800 A2

ii. Amount on food = $\frac{1}{4} \times$ GH₵4,800 = GH₵1,200 M1 A1

iii.. Amount on food = $\frac{x}{16} = \frac{\text{GH₵4800}}{16} =$ GH₵300 M1 A1

iv..His salary left as fraction of the his total monthly income = $\frac{\text{GH₵2100}}{\text{GH₵4800}} = \frac{7}{4}$ A2

5. U = {18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 30, 31, 32, 33, 34, 35, 36} M1

P = {18, 21, 24, 27, 30, 33, 36} M1

Q = {18, 24, 36}

R = {18, 20, 22, 24, 26, 28, 30, 32, 34, 36} A1

b. (i) $P \cap Q$ = {18, 24, 36} A1

(ii) $Q \cap R$ = {18, 24, 36} A1

(iii) $P' \cap R$ = {19, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 35} ∩ {18, 20, 22, 24, 26, 28, 30, 32, 34, 36} M1

= {20, 22, 26, 28, 32, 34} A1

(v) $P \cap Q = Q \cap R$

A1

c. (i) Length ,L = $(2x + 5)$m

Width, W = $(x - 10)$m

Perimeter = 80m

Perimeter of a rectangle = 2L + 2W M1

∴ 2L + 2W = 80 M1

$2(2x + 5) + 2(x - 10) = 80$ M0.5

$4x + 10 + 2x - 20 = 80$ M0.5

$6x - 10 = 80$ M0.5

$6x = 80 + 10$ M0.5

$x = \frac{90}{6} = 15$ A1

(ii) Length = $2 \times 15 + 5 = 35$m M0.5

Width = $15 - 10 = 5$m M0.5

Area = Length × Width

$= 35 \times 5 = 175\text{m}^2$ A1

6. a. for $x = -2$

$y = 5 - 2(-2)$

$y = 9$

for $x = -1$

$y = 5 - 2(-1)$

$y = 7$

for $x = 1$

$y = 5 - 2(1)$

$y = 3$

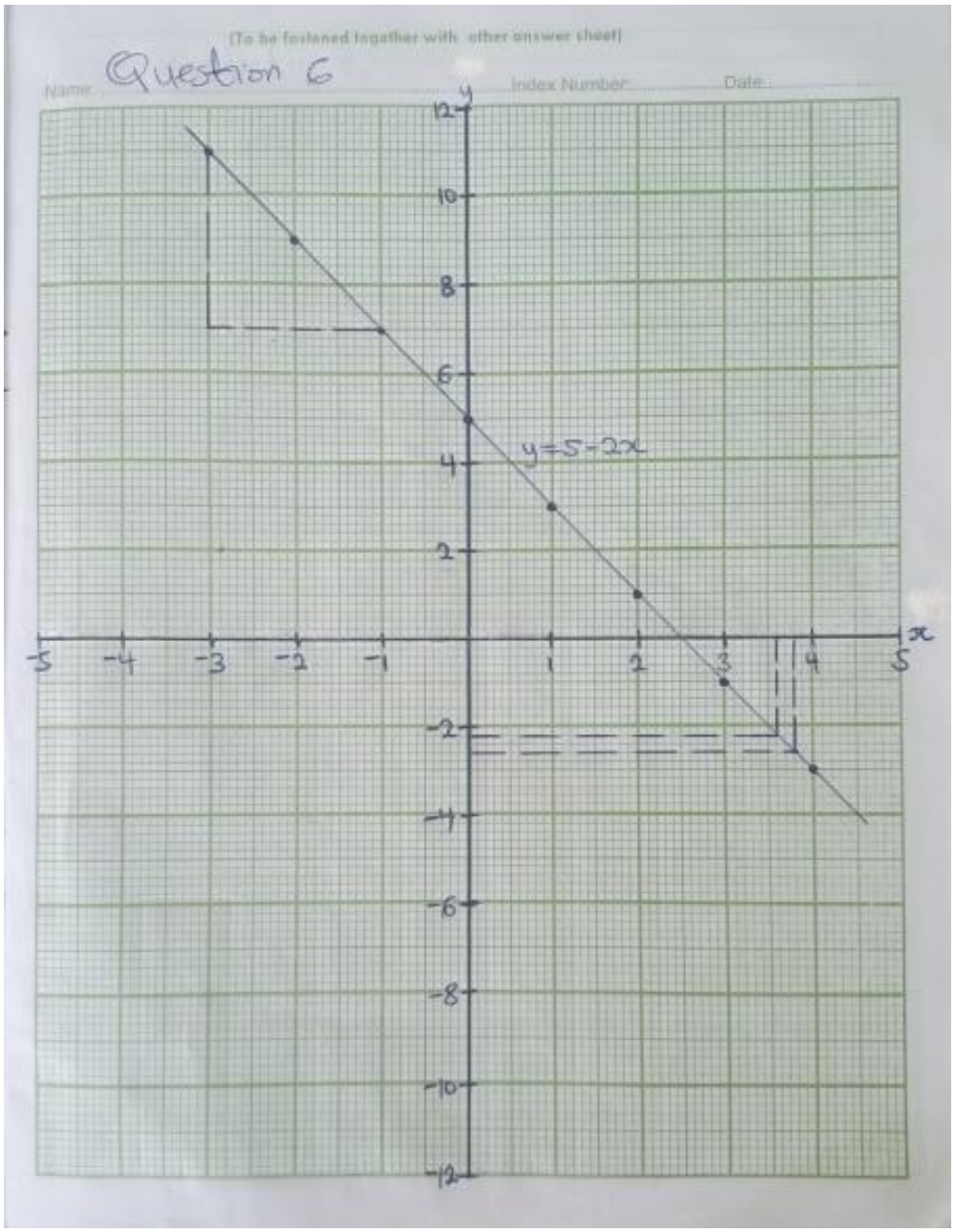
for $x = 3$

$y = 5 - 2(3)$

$y = -1$

x	-3	-2	-1	0	1	2	3	4
y	11	9	7	5	3	1	-1	-3

4 marks for table



2 marks for axes @ 1each

4 mark for points @ $\frac{1}{2}$ each

1 mark for straight line

d.. from the graph;

i. when $x = 3.6, y = -2.2$ A1

ii. when $y = -2.6, x = 3.8$ A1

ii. Gradient= $\frac{y_2-y_1}{x_2-x_1} = \frac{7-11}{-1--3} = \frac{-4}{1} = -4$ M1A1